

Manasvini Kandikonda

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About

AI & Data Science undergraduate with hands-on experience building end-to-end ML, computer vision, and LLM-based systems. Worked on real-time vision control, medical RAG applications, and predictive modeling projects. Interested in applying AI to interdisciplinary domains, and large-scale engineering problems.

Education

B.Tech in Artificial Intelligence and Data Science

2024 – 2028

Amrita Vishwa Vidyapeetham

CGPA: 8.56

Higher Secondary Education

CISCE — New Horizon Public School, Bengaluru

Percentage: 95%

Skills

Programming: Python, Java, C, C++

AI & ML: Machine Learning, Deep Learning, Computer Vision, Transformers, Neural Networks

Backend: REST APIs, FastAPI, Node.js, MongoDB, SQLite

DevOps and Tools: Git, Docker, Linux, AWS (S3, Lambda)

Web: HTML, CSS, JavaScript, React

Experience

Intern — 314e

December 2025 – January 2026

- Built a Retrieval-Augmented Generation (RAG) system for querying patient medical records
- Implemented semantic chunk retrieval with source-page highlighting for explainability
- Worked with transformer-based embeddings and vector search to reduce hallucinations

End-to-End Deployed Projects

Psychoactive Drug Prediction System - GitHub

- Built a fusion machine learning framework combining chemical descriptors and behavioral data.
- Developed regression models to predict Impulsivity and Sensation Seeking scores.
- Designed a classification pipeline to distinguish psychoactive vs non-psychoactive substances.

Facial Emotion Detection: ML vs DL comparison - GitHub

- Conducted a comparison of classical ML and deep learning models for facial emotion recognition on the KDEF dataset, evaluating accuracy, confusion matrices, and class-wise performance across 7 emotions.
- Implemented multiple ML pipelines such as HOG, Local Binary Pattern, Logistic Regression, Simple Vector Machine, Random Forest and Deep Learning architectures including ConvNeXt-Tiny, ResNet50, EfficientNet, Vision Transformer.
- Observed significant performance gains with deep learning models, achieving up to 95% accuracy with Vision Transformers.

LLM-Based Optimized Synthesis of Gold Nanoparticles for Cancer Treatment - GitHub

- Created a custom dataset created using pre-trained transformers with 20,000 nanoparticles and 26 features.
- Results showcased the optimal synthesis conditions, including a detailed review of the nanoparticles' compatibility.

Remote Control Car Vision & Control System - GitHub

- Developed a remote-controlled car with speed control, camera feed, Bluetooth modules, and obstacle detection.

- Implemented **shortest-path finding and tracking** using **Computer Vision** based techniques.

Safe Chat — GitHub

- Designed a lightweight QUIC-based chat system over HTTP/3 with end-to-end classical and quantum encryption.

Positions of Responsibility

Vice Chair — IEEE Women in Engineering (WIE)

July 2025 – Present

- Organized technical talks featuring professionals from Oracle and Schneider Electric.
- Led and coordinated over **five** technical and outreach events with 100+ participants during the 2025 odd semester.

Captain - CISCE Karnataka Basketball Team

2019

- Led a State Basketball Team to a Gold medal at the Sub Junior National Basketball tournament.

Publications

- *The Effects and Classification of Psychoactive Substances: A Machine Learning Approach*. CSITSS 2025 — Paper
- *Contour Based Tracking vs. Hough Transform Based Tracking: A RC Car Experiment*. WCONF 2025 — Paper

Hackathons

- **Finalist** at the Novo Nordisk Hackathon 2025.
- **Runners-Up** at an electronics-based hackathon (2025).
- Created a medical drug recommendation chatbot using LLaMA-3.1 (2024) - Github

Extracurriculars

- Two-time Karnataka State Basketball player and Captain of the CISCE Karnataka Team.
- Active member of the University Basketball Team (2024 – Present).
- Outreach: Developed a mental health application under a UN initiative and mentored school students in app development